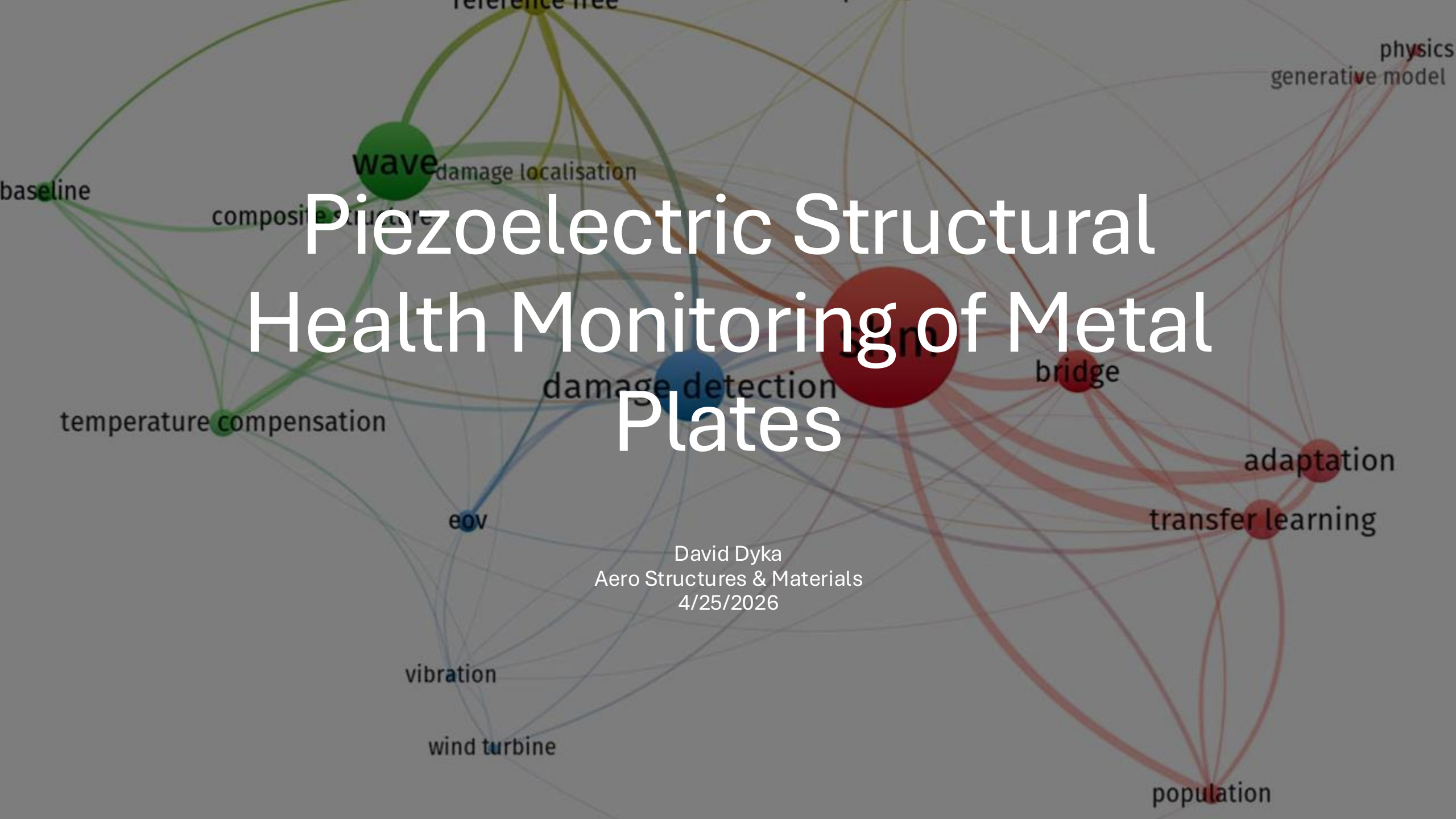




Piezoelectric Structural Health Monitoring of Metal Plates

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Aero Structures & Materials
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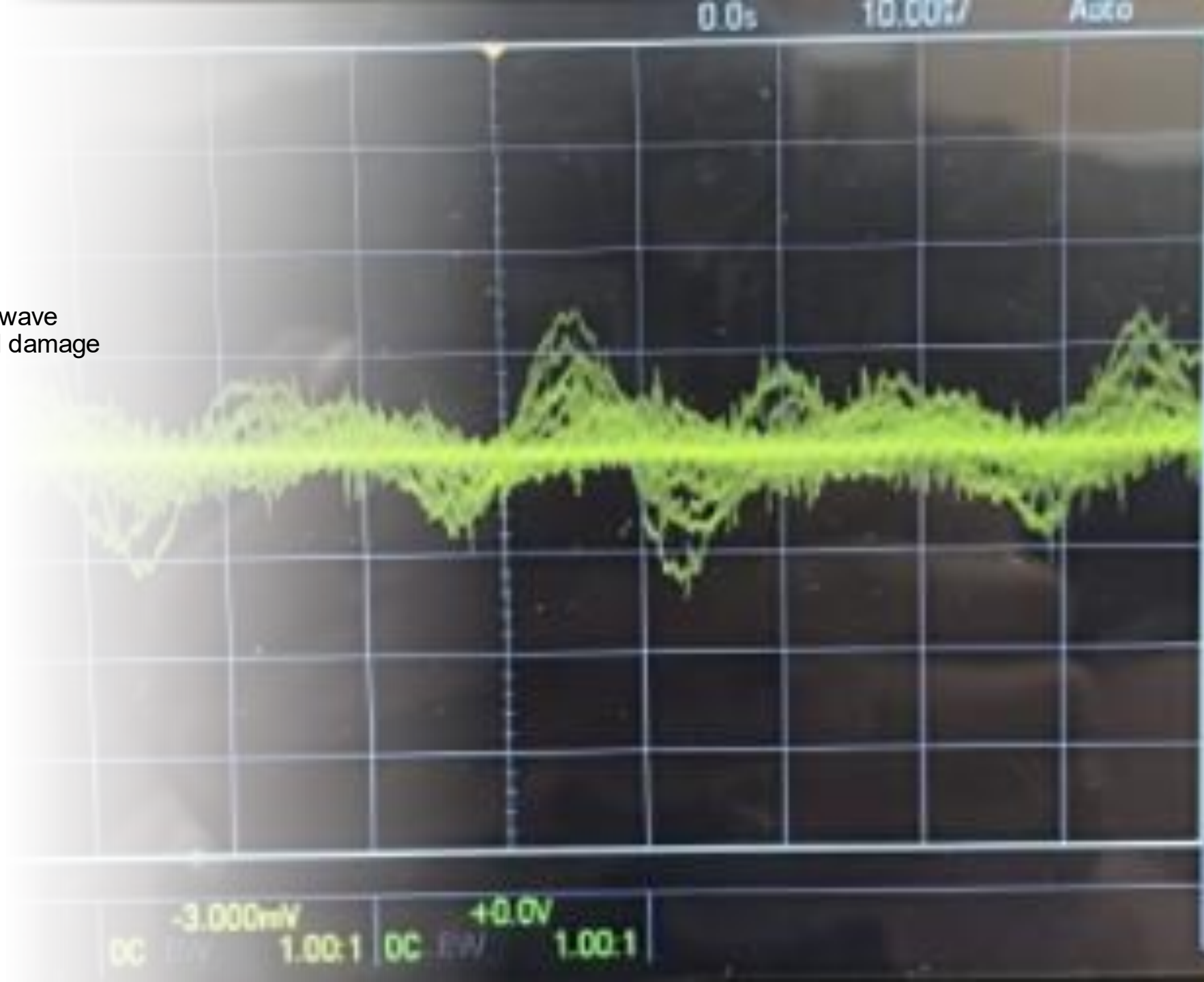
Motivation

A close-up, low-angle view of an aircraft's wing and engine against a sunset sky. The wing is dark and metallic, with a large engine mounted below it. The sky is a mix of orange, yellow, and blue, with the sun low on the horizon. The aircraft's tail fin is visible in the upper left corner.

- Structural damage in aircraft can be difficult to detect
- Small cracks can grow and lead to failure
- Traditional inspections are periodic and manual
- Structural Health Monitoring (SHM) enables continuous monitoring

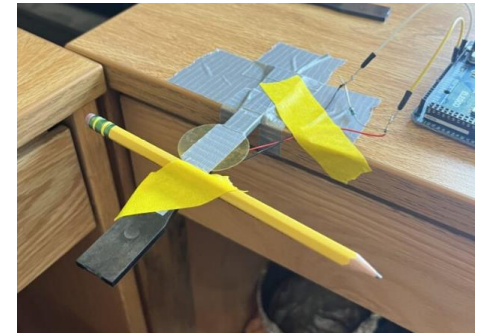
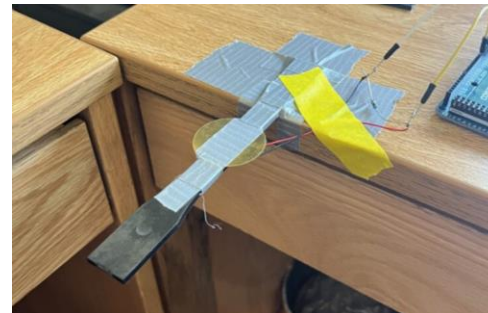
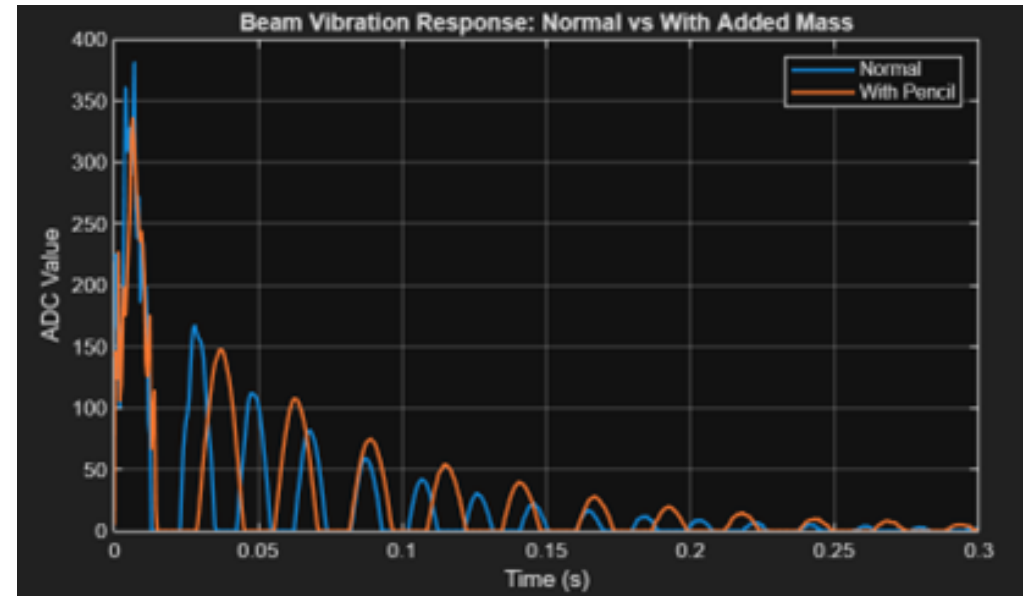
Objective

- Investigate whether piezoelectric wave transmission can detect structural damage
- Compare different damage levels:
 - Undamaged
 - Scratch
 - Cut
 - Severe damage
- Evaluate signal-based features:
 - Energy
 - Peak amplitude
 - RMS



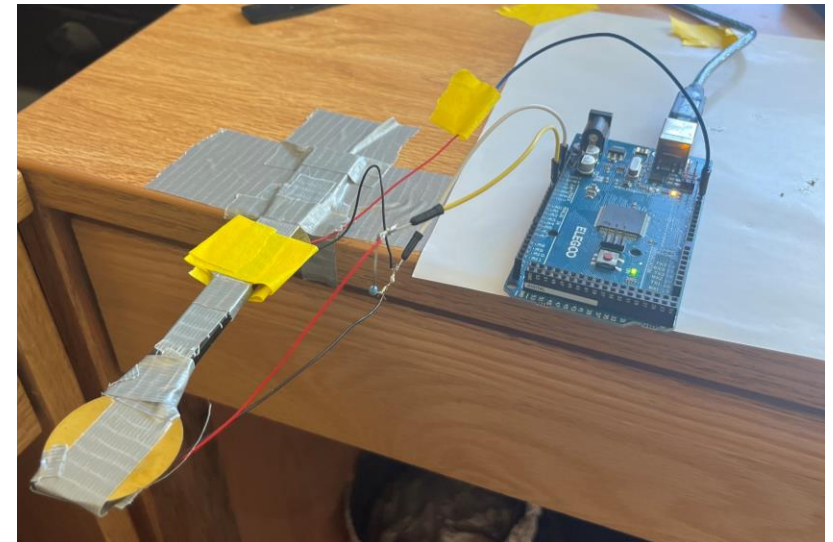
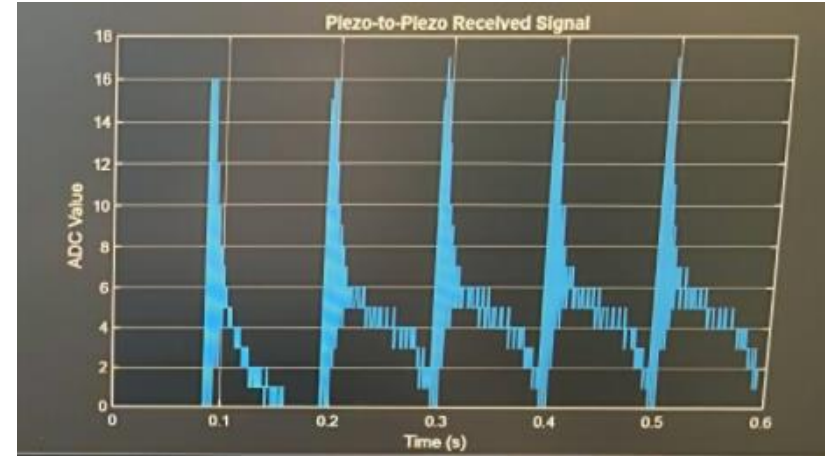
Early Experiment: ABS Beam With Mass Drop

- ABS beam with piezo sensor near taped end
- Dropped small weight at same height to generate signals
- Measured response at receiver and demonstrated that piezo sensors can detect vibration signals
- Added a mass (pencil) and observed a phase shift in the vibration response graph



Early Experiment: ABS Beam With Pulses

- Transitioned to Arduino-based pulse generation
- Sent signals through piezo actuator
- Measured transmitted response
- Improved repeatability but limited signal quality and frequency control



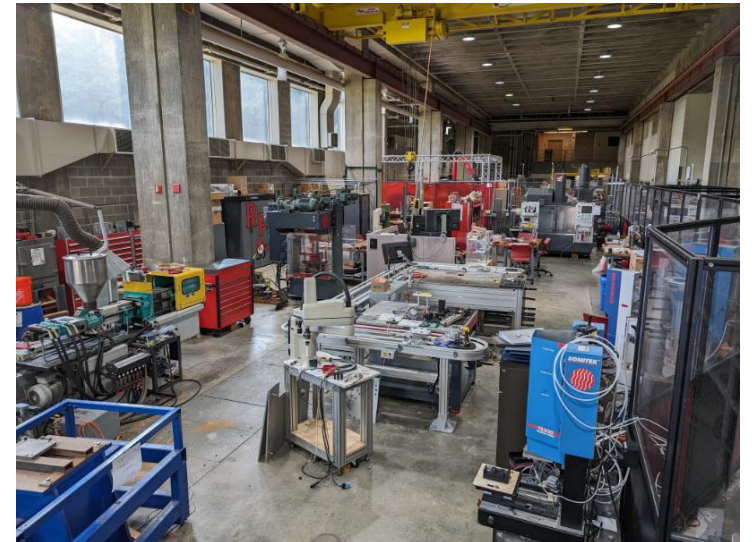
Manufacturing and Transition to Lab Equipment

Manufacturing

- Cut aluminum plates from scrap using sheet metal tools
- Prepared piezo sensors:
 - Soldered resistors
 - Tinned wires for connections
- Epoxied piezos to plates and cured overnight

Lab Integration

- Learned to use:
 - Function generator (wave input)
 - Oscilloscope (signal measurement)
- Set up full system:
 - TX (actuator) → RX (sensor)
- Tuned frequency and waveform parameters



Experimental Setup

- Piezoelectric actuator (TX) sends wave
- Piezoelectric sensor (RX) receives signal
- Metal plate as test structure
- Function generator:
 - 50 kHz sine burst
 - 1000 cycles
 - 0.2 second period
- Oscilloscope records response and saves on a CSV with 5ms window

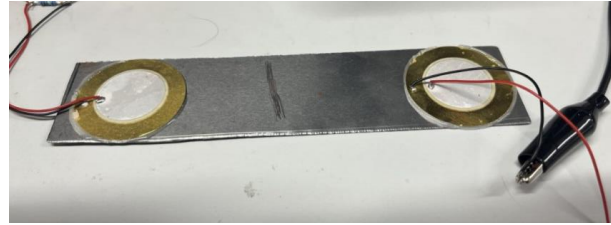


Damage Cases

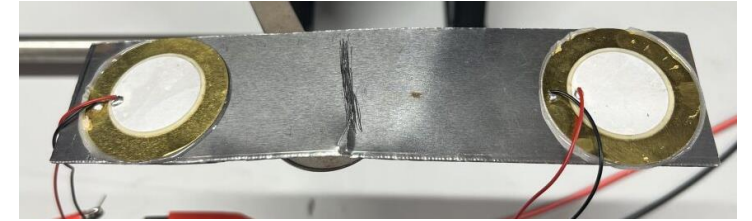
Undamaged plate (baseline)



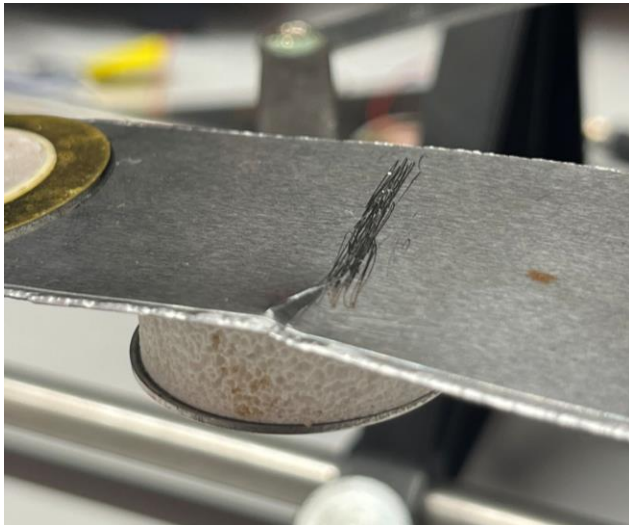
Light surface scratch



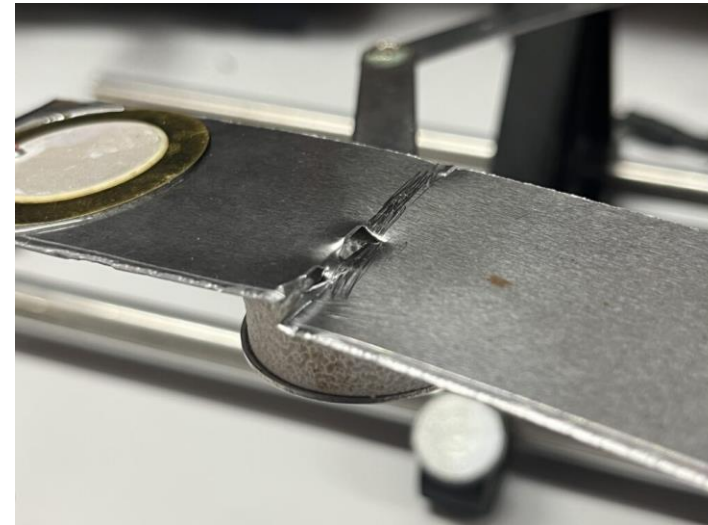
Heavy surface scratch



Single cut (moderate damage)



Multiple cuts (severe damage)



Signal Processing

- Removed DC offset
- Aligned signals using cross-correlation
- Extracted features:
 - Energy
 - Peak amplitude
 - RMS
- Removed outliers statistically

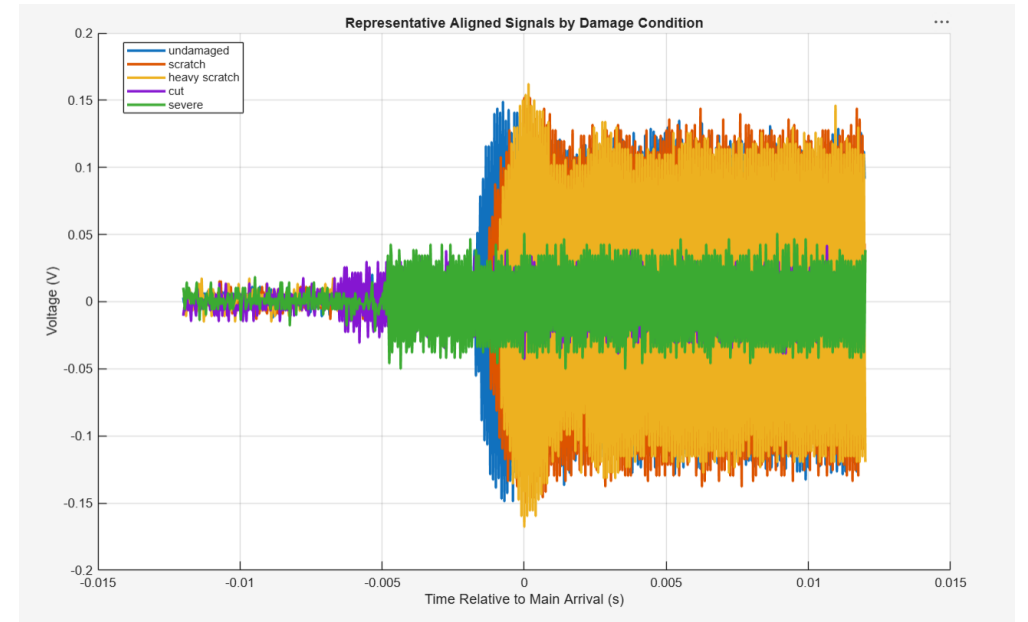
```
% Remove DC offset
v = v - mean(v);

% Align signal to baseline using cross-correlation
[c,lags] = xcorr(v, ref);
[~,idx] = max(abs(c));
shift = lags(idx);
v_aligned = circshift(v, -shift);

% Extract features
energy = trapz(t, v_aligned.^2);
peak_amp = max(abs(v_aligned));
rms_val = sqrt(mean(v_aligned.^2));

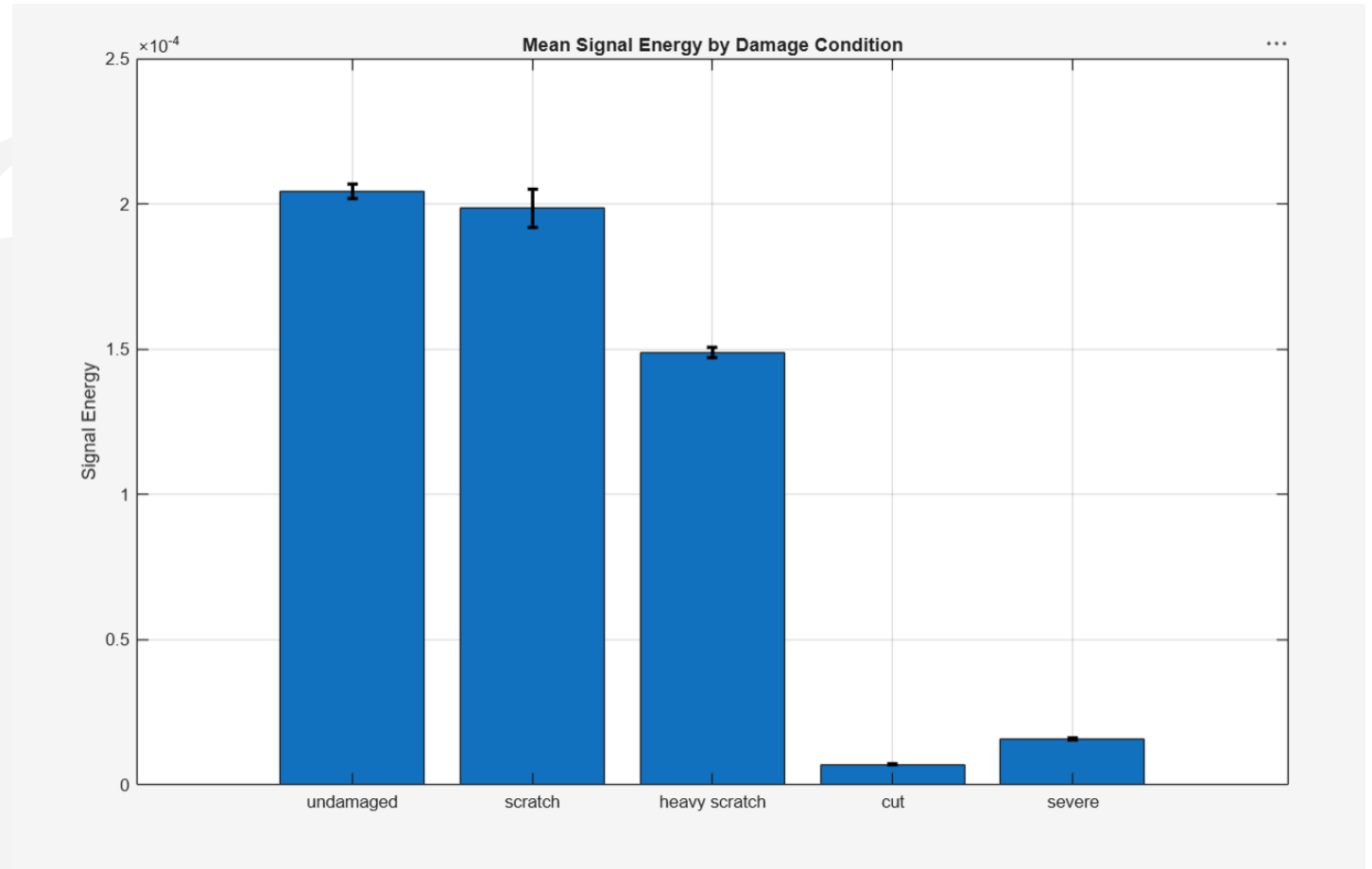
% Remove outliers using z-score
z = (energy - mean(energy_all)) / std(energy_all);
valid = abs(z) < 1.5;
```

Aligned Signals



Signal Energy Results

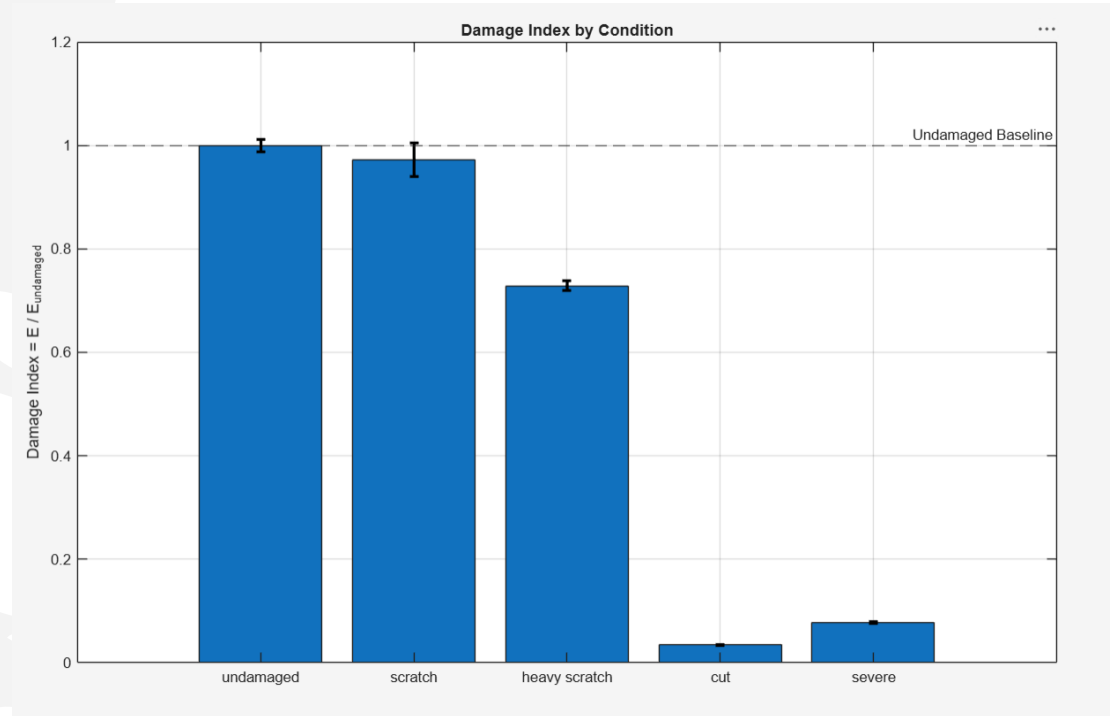
- Undamaged and scratch conditions show nearly identical energy levels
- Significant reduction observed for cut and severe damage
- Energy provides clear separation between damage states
- Heavy scratch produces a moderate reduction in energy, bridging the gap between surface damage and structural damage.



Damage Progression

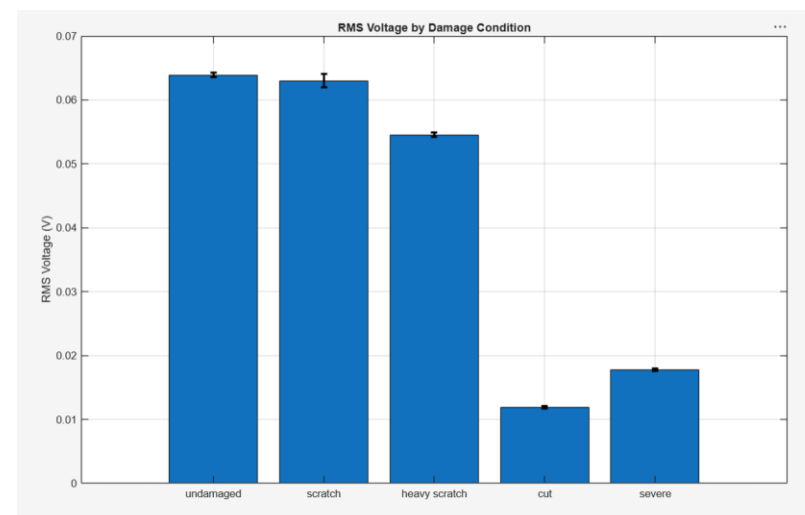
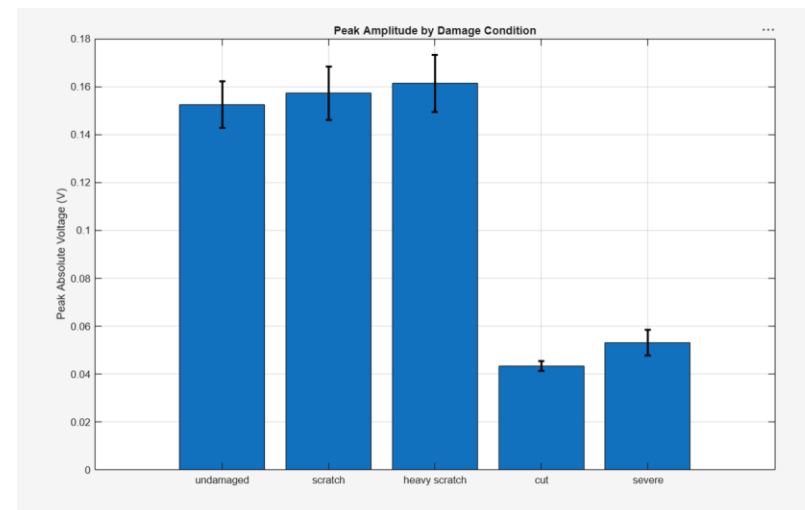
$$\text{Damage Index (DI)} = \frac{E}{E_{\text{undamaged}}}$$

- Light scratch: minimal change (~3%)
- Heavy scratch: moderate reduction (~27%)
- Cut: major attenuation (~96%)
- Double cut: strong attenuation (~92%)
- A clear detection threshold exists between surface damage and structural damage



Peak & RMS Results

- Both metrics show reduced values for damaged conditions
- Trends are consistent with energy-based analysis
- Confirms reliability of signal-based detection



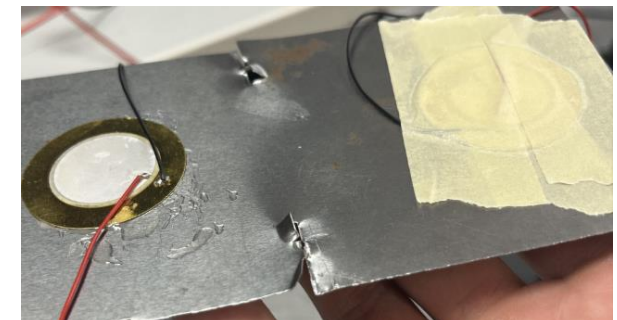
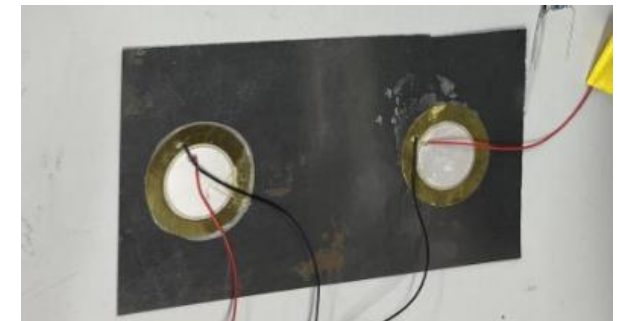
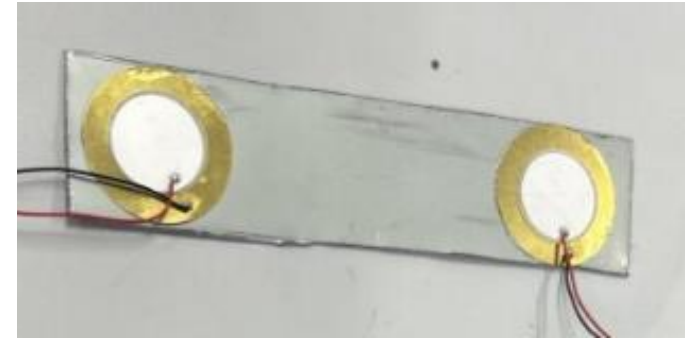


Key Observations

- Light scratches are not detectable using this method
- Increasing surface damage produces gradual signal reduction
- Structural damage (cuts) causes large energy loss
- Detection behavior is nonlinear and depends on wave path
- The single-cut case produced lower transmitted energy than the double-cut case, indicating that wave attenuation depends on damage location relative to the primary propagation path.

Additional Experiments

- Conducted additional tests on:
 - Small thick plate
 - Large thin plate
- Explored how structural geometry and thickness influence wave propagation
- Observed variations in:
 - Signal amplitude
 - Wave attenuation behavior
- Results suggest that SHM response depends on material geometry in addition to damage

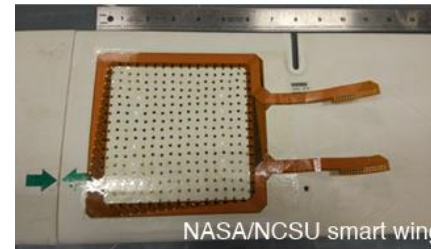
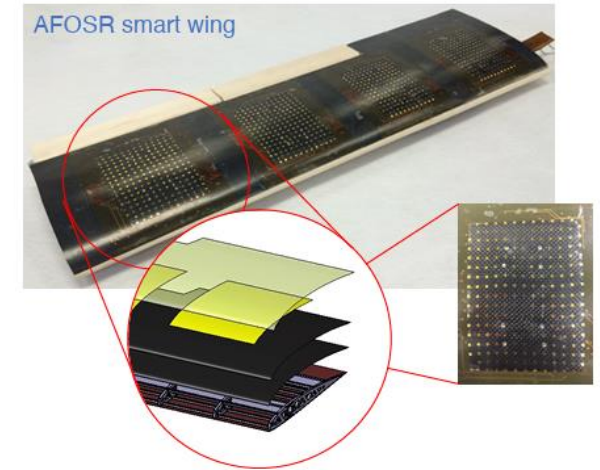
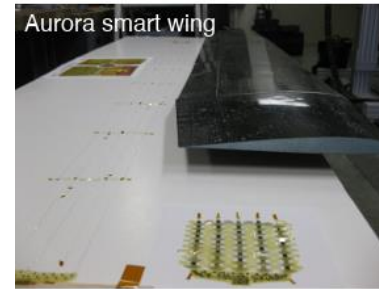


Conclusions

- Piezoelectric SHM can detect structural damage
- Energy is the most reliable metric
- Small surface damage is not detectable with this setup
- Larger damage produces clear signal attenuation
- Signal response is not strictly proportional to damage severity

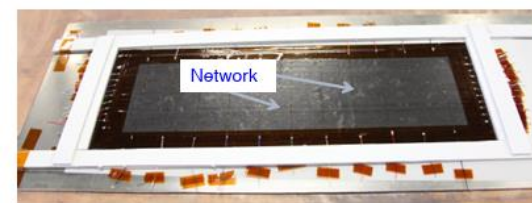
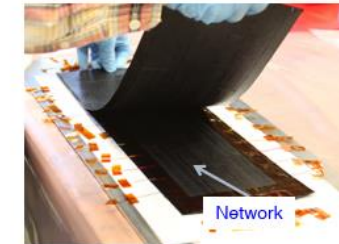
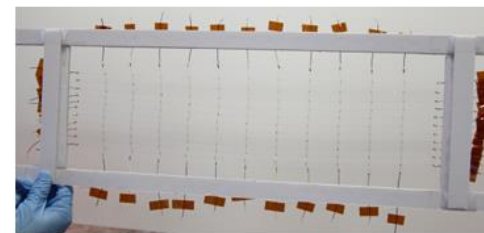
Future Work

- Test different frequencies
- Use multiple sensors
- Improve sensor coupling
- Apply advanced signal processing
- Sensor Integration into Composites
- Apply system to RC aircraft for real-world testing



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Sensor Integration into Composites



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